



Pisa University

LIGHTGBM

A HIGHLY EFFICIENT GRADIENT BOOSTING DECISION TREE

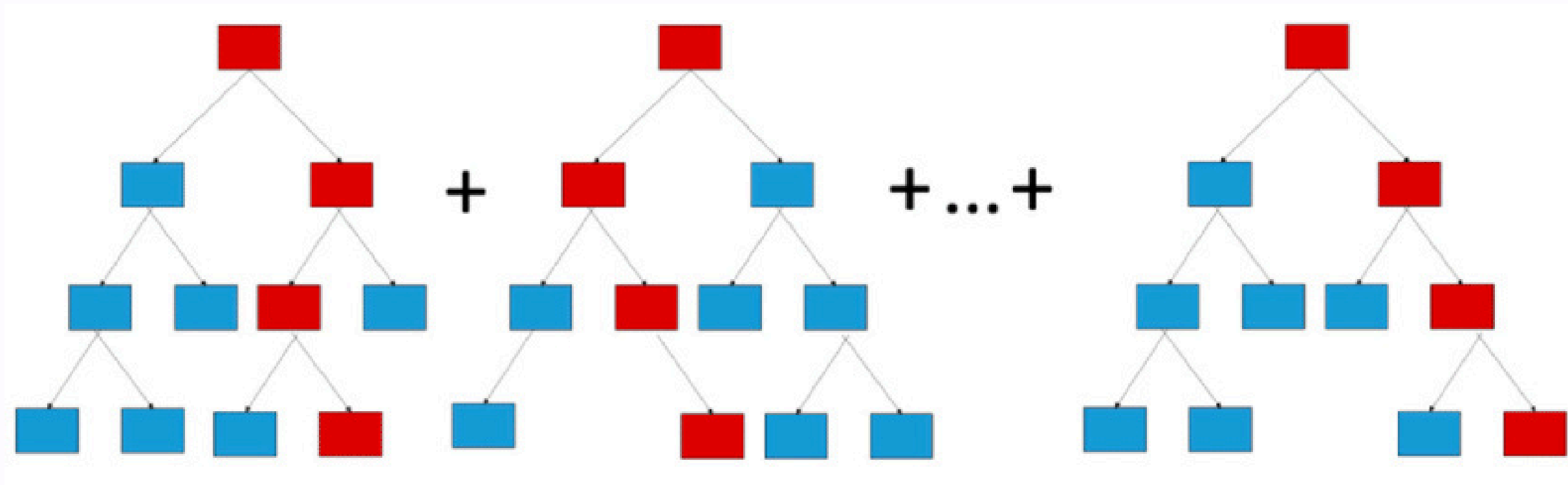
Presented by Carmine Vitiello

Information Retrieval | 2026





ABSTRACT



Gradient Boosting Decision Tree (GBDT) has many implementations, but the efficiency and scalability are still unsatisfactory. Therefore two novel techniques were proposed: *Gradient-based One-Side Sampling* (GOSS) and *Exclusive Feature Bundling* (EFB)





OVERVIEW

Introduction

Problems

Metric GOSS o1

Metric GOSS o2

Theoreme GOSS

Theoretical EFB

EFB Algorithm

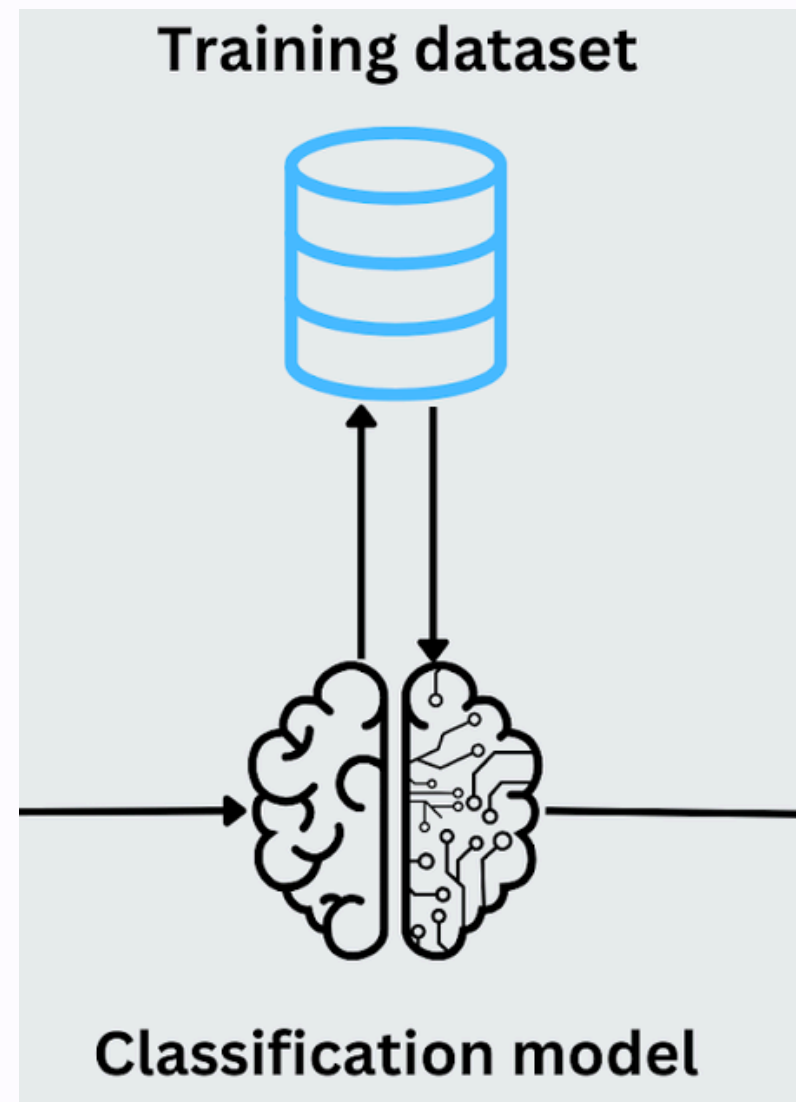
Result

Conclusion





INTRODUCTION



GBDT is an algorithm that can be used to solve many different types of tasks (classification, learning to rank, click prediction, etc.), but it has some performance issues when working with big data.

To fix this issue, two algorithms were introduced: GOSS and EFB. The first one is used for sampling the data through the gradient, because data instances with different residuals affect the computation of information gain differently compared to random sampling. The second one aims to reduce the number of effective features, specifically in a sparse feature space.





PROBLEMS

01

The conventional implementations of GBDT require all the data instances to be scanned for every feature in order to estimate the information gain of all the possible split points. Consequently, the processing difficulty and the duration will be elevated when managing substantial volumes of data.

02

For reducing the number of data instances and features is not a trivial task, so it is necessary to sample the data before the training operation, because they should have a different weight than the absolute residual.

03

The bundling of the features is a very hard task, because it requires combining exclusive features while avoiding the exclusion of any with high relevance in the process. The problem is NP-hard, so it needs to be reduced to a known problem such as graph coloring.





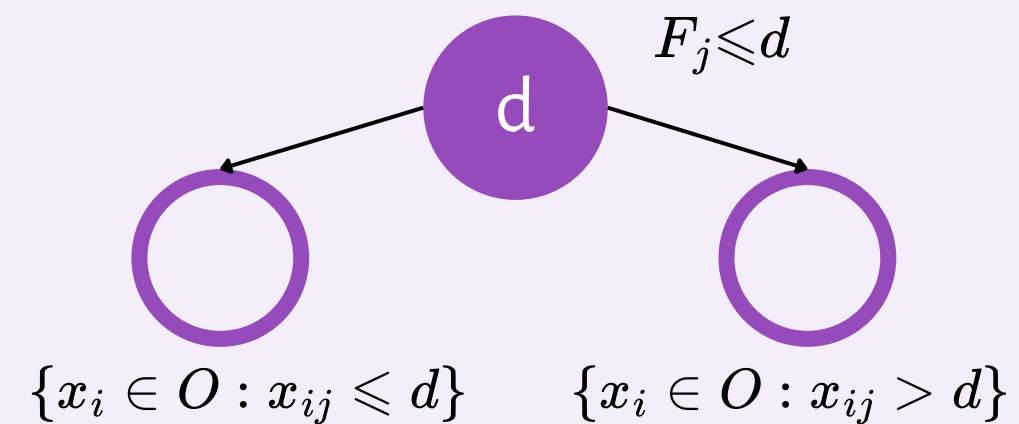
METRICS GOSS 01

THEORY

$$V_{j|O|}(d) = \frac{1}{n_O} \left(\frac{(\sum_l g_i)^2}{n_{l|O|}^j(d)} + \frac{(\sum_r g_i)^2}{n_{r|O|}^j(d)} \right)$$

The algorithm splits each node based on the largest information gain, selecting the most informative feature. The information gain is usually measured by the variance after the split.

DRAW



This is standard variance on the splitting of the single node **d** on the feature **j**, where **nO** is the number of instances in the dataset and **gi** are the gradients of the loss function of the model.





METRIC GOSS 02

PROCESS

In this case, the training instances were sorted according to the absolute values of their gradients in descending order. Therefore, they were split into subsets A and B of the original training dataset, where A contains the top-a instances and B contains a random-b samples taken from the complementary subset of A.

FORMULA

$$V'_j(d) = \frac{1}{n} \left(\frac{(\sum_{A_l} g_i + \frac{1-a}{b} \sum_{B_l} g_i)^2}{n_l^j(d)} + \frac{(\sum_{A_r} g_i + \frac{1-a}{b} \sum_{B_r} g_i)^2}{n_r^j(d)} \right)$$





THEOREME GOSS

ERROR 01

The algorithm splits each node by selecting the feature with the largest information gain, that is, the most informative feature.

$$\varepsilon(d) = |V'_j(d) - V_j(d)|$$

This is the computing the approximation error in GOSS as ε of node d with probability at least of $1 - \delta$

ERROR 02

The asymptotic approximation of GOSS for V'_j

$$O\left(\frac{1}{n^{j_l}(d)} + \frac{1}{n^{j_r}(d)} + \frac{1}{\sqrt[2]{n}}\right)$$

If the split is not too unbalance the approximation error will be dominated by the second term which decreases to zero in $O(\sqrt[2]{n})$ with $n \rightarrow \infty$

$$\varepsilon(d) \leq C_{a,b}^2 * \ln \frac{1}{\delta} * \max\left\{\frac{1}{n_l^j(d)}, \frac{1}{n_r^j(d)}\right\} + 2DC_{a,b} \sqrt{\frac{\ln 1/\delta}{n}}$$





THEORETICAL EFB

THEORY

To reduce the number of the features, it's possible to reduce the problem to graph coloring problem with an instance $G = (V, E)$

The proposed algorithm splits the incident matrix of G in $|V|$ features (the nodes of the graph) and merges the node with the same color into the corresponding exclusive bundle.

RESULT

The splitting operation provides an approximation, because it's possible to find some conflicts in the bundles. However, these conflicting groups will affect the training by at most

$$O\left([(1 - \gamma)n]^{-2 \div 3}\right)$$

where γ is the maximal conflict rate. The performance of the model change from **$O(\text{\#data} \times \text{\#feature})$** to **$O(\text{\#data} \times \text{\#bundle})$** , where **$\text{\#bundle} \ll \text{\#feature}$** .





EFB ALGORITHM

ALGORITHM 01

1. Construct the graph G
2. Sort the graph for weighted edges, then there is more possible to fail earlier the resolution of the coloring graph problem
3. Check if it's possible to add the feature in the already existing bundle
 - a. It's possible if there aren't K bundles that already had it
4. Otherwise make a new one

ALGORITHM 02

1. Compute the total of the Bin with respective ranges
2. Merge the bin with the specific offset

Example

Exclusive Features: $\{F1, F3, F7\}$

F1: $[0 - 9]$

F3: $[10 - 19]$

F7: $[20 - 29]$

The new **Bin** will have a single range from 0 to 29





RESULT

Accuracy	xgb_exa	xgb_his	lgb_baseline	SGB	Light GBM
LETOR	0.4977	0.4982	0.5277	0.5239 ±6e-4	0.5275 ±5e-4
KDD10	0.7796	OOM	0.7873	0.7759 ±3e-4	0.7873 ±1e-4

Training Seconds Cost	xgb_e xa	xgb_ his	lgb_baseline	EFB_only	Light GBM
LETOR	5.55	0.63	0.49	0.46	0.31
KDD10	108.27	OOM	39.85	6.33	2.85

It's used a Server Linux with
a. two E5-2670 v3 CPUs
b. 256 GB memories

Name	#data	#feature	Description	Task	Metric
LETOR	2M	136	Dense	Ranking	NDCG
KDD10	19M	29M	Sparse	Binary Classification	AUC





CONCLUSION

During the final phases, a comparison was made between GOSS and EFB algorithms to determinate thier influence during the experiments. GOSS was compared to SGB with differents percentage of sampled data, and there were no fluctuations in the accurancy for GOSS, which perfomed better than SGB.

The EFB algorithm was set a zero conflicts maximum rate, and it contributed positively to the training process. It can merge a large number of the sparse features without the additional cost of maintaining a nonzero data table for each feature during tree learning.

The results confirm that LightGBM model with GOSS and EFB improvements can significantly outperform XGBoost and SGB in term of computational speed and memory consumption.





REFERENCE





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THANK YOU

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